

Project Demonstration

Date	4 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Project Demonstration Overview

This section demonstrates the complete workflow of the **Voice-Based Concept Understanding Analyzer**. The application accepts a student's voice explanation, converts speech into text, performs semantic analysis, evaluates conceptual understanding, and generates a detailed PDF report.

1. System Architecture

DATA FLOW DIAGRAM (LEVEL 0)

Voice-Based Concept Understanding Analyzer

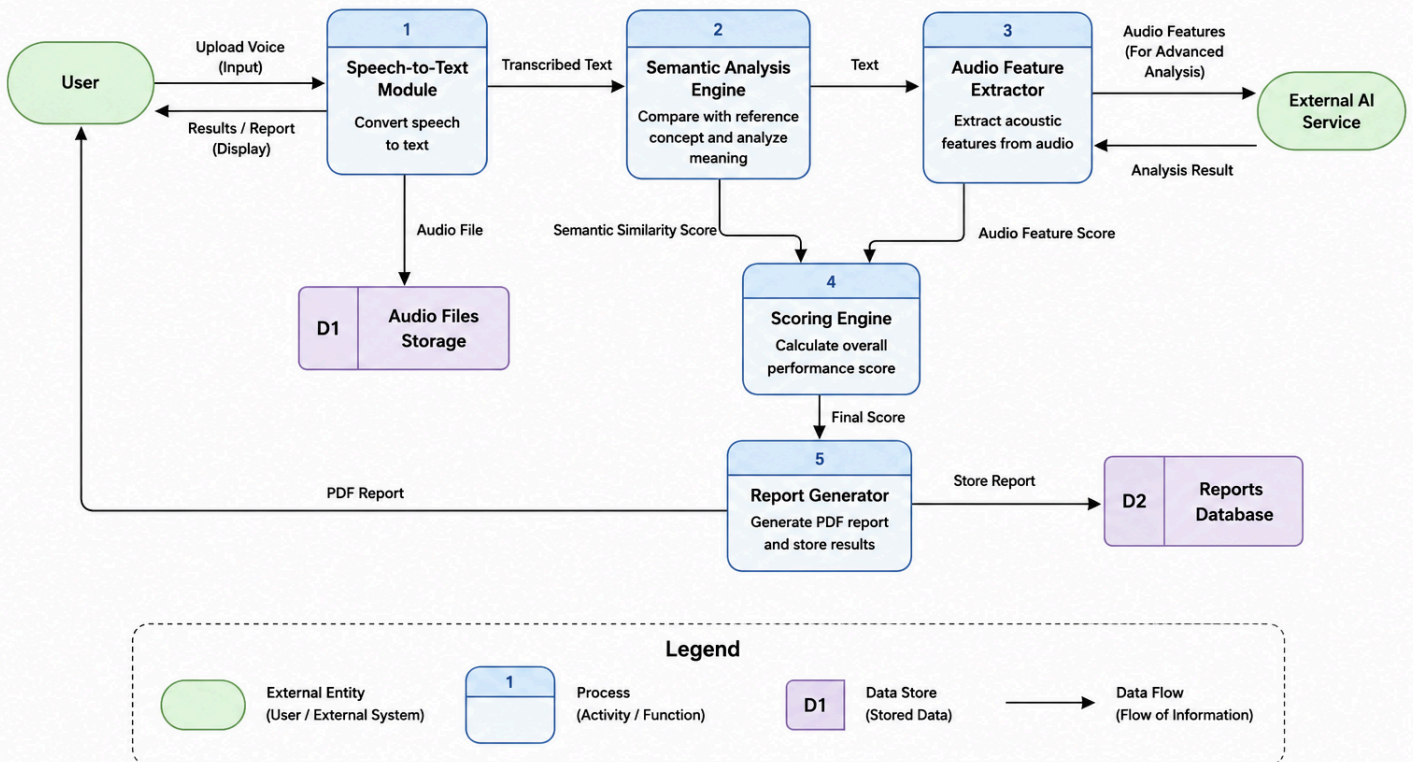
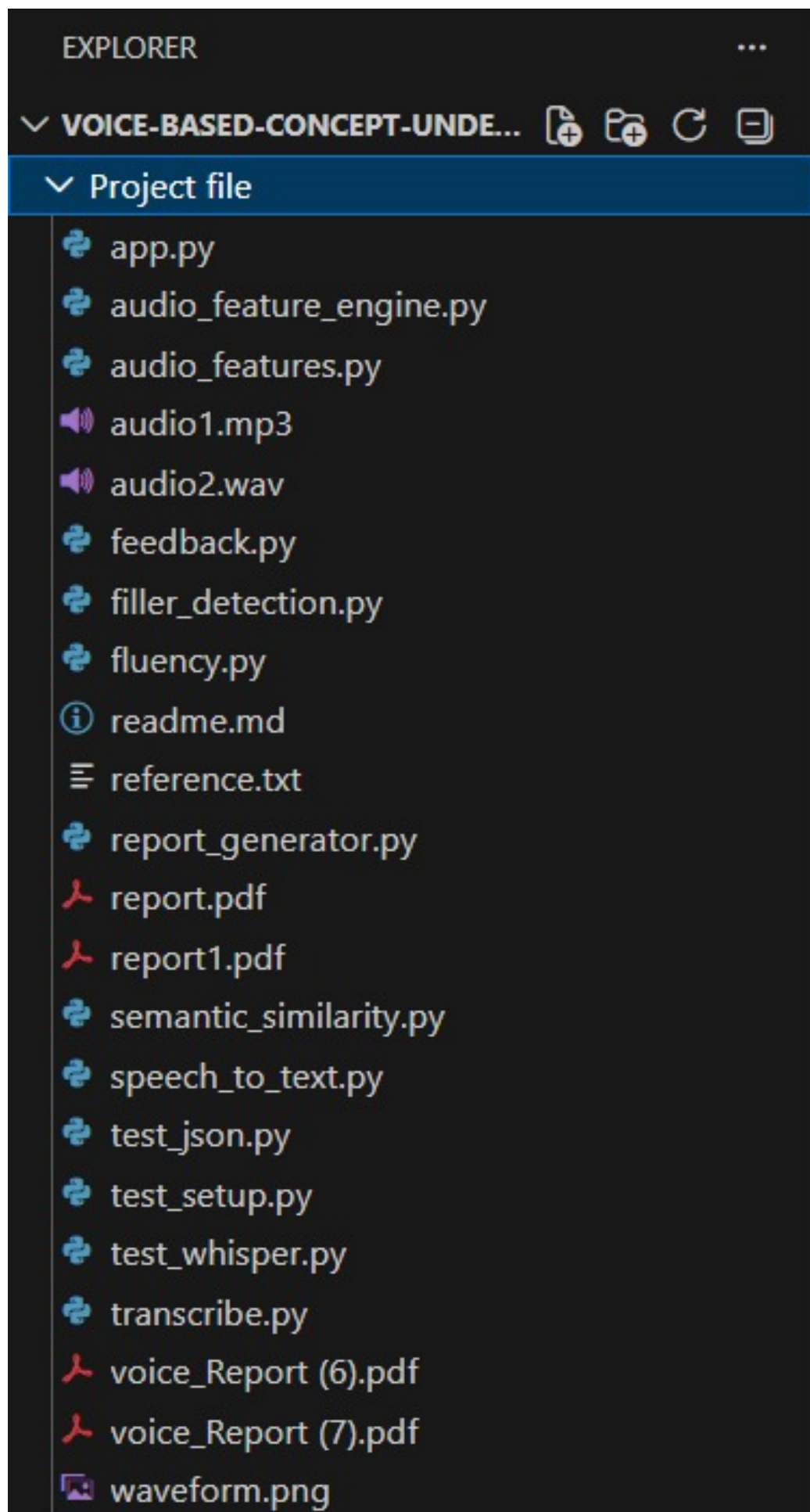


Figure 1: Data Flow Diagram

2. Project Structure



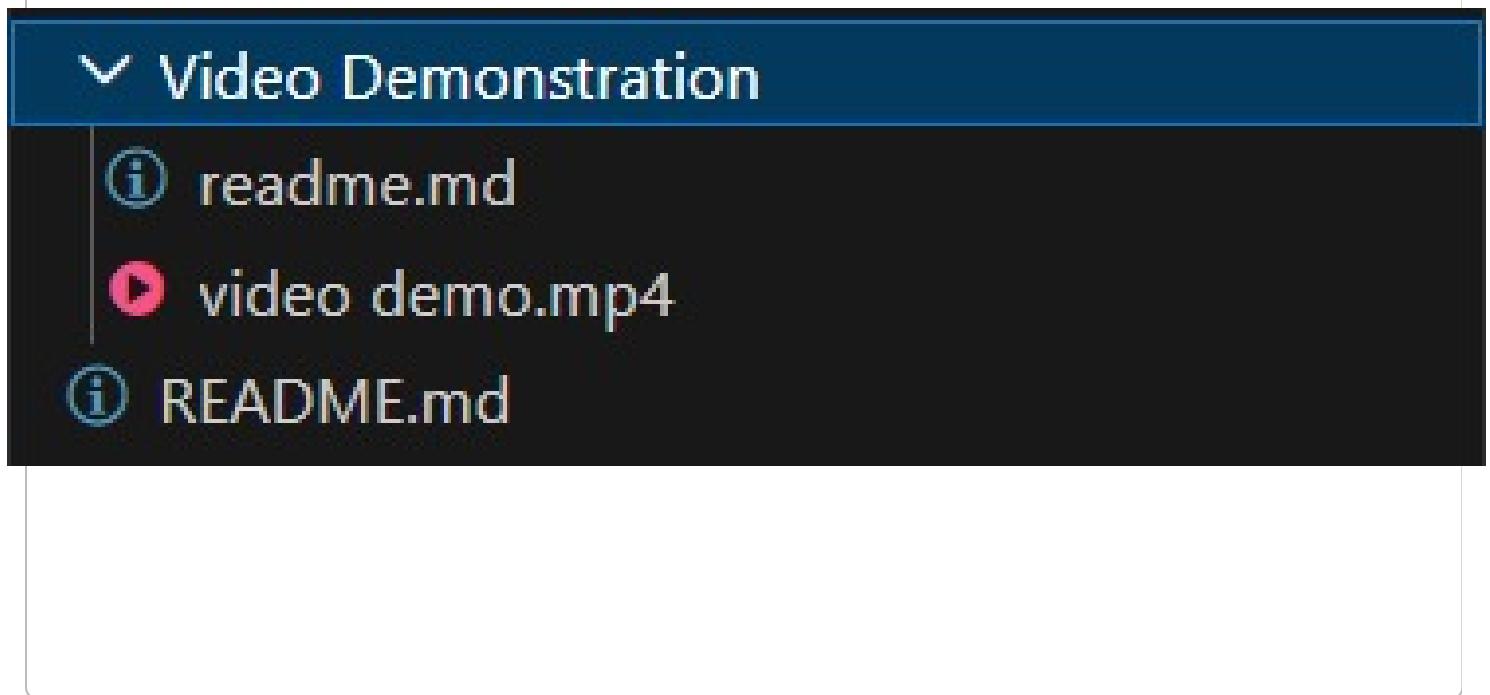


Figure 2: Project Folder Structure in VS Code

3. Running the Application

```
PS D:\Voice Based Concept Understanding Analysis\Source_Code> py -m streamlit run app.py
2026-07-05 09:38:17.995 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.43.185:8501
```

Figure 3: Streamlit Server Started Successfully

4. Streamlit Application Running

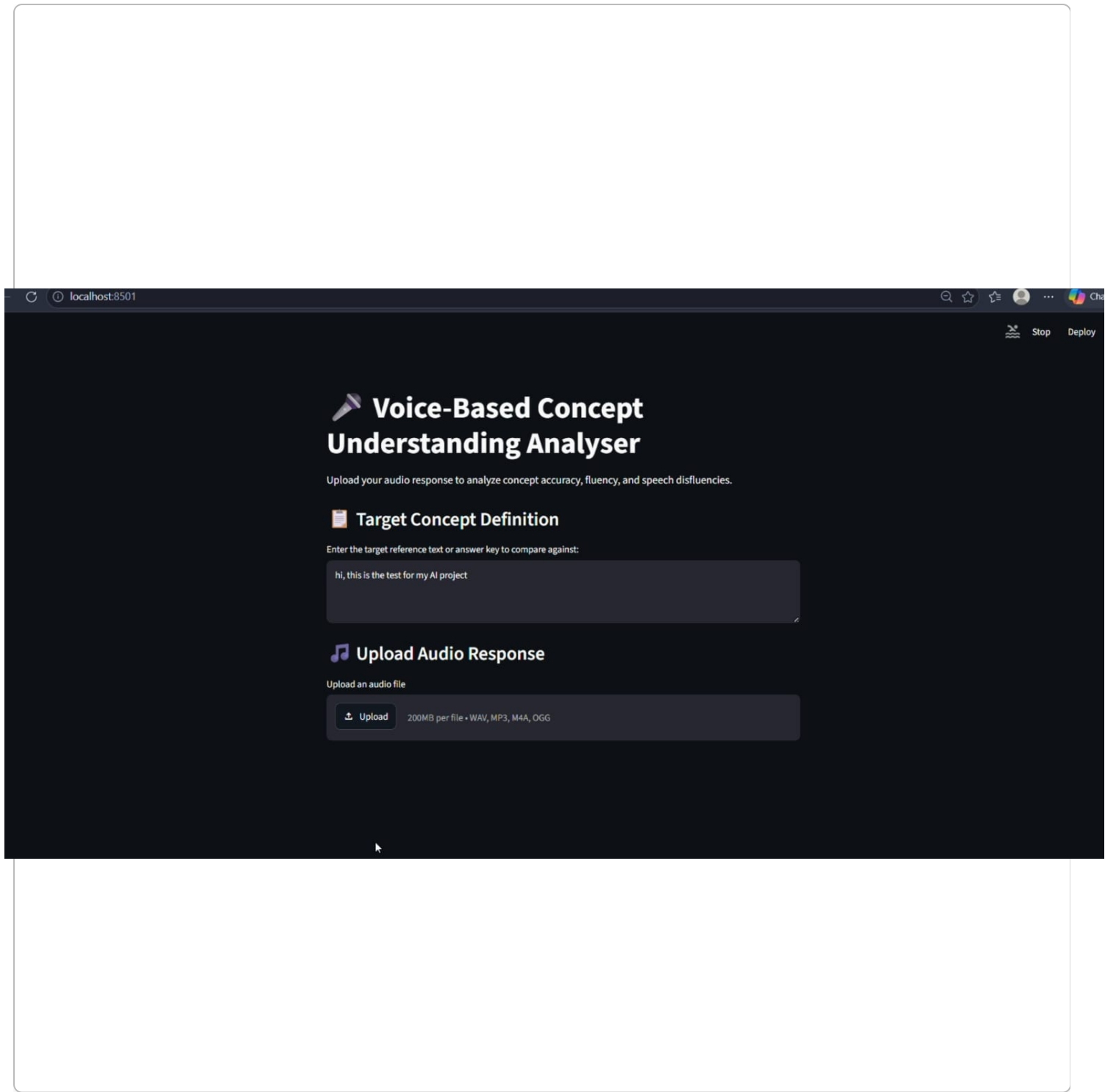


Figure 4: Application Running on Localhost

5. Application Workflow

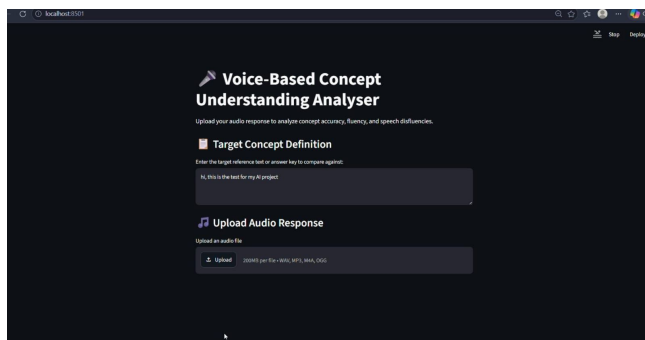


Figure 5: Home Page

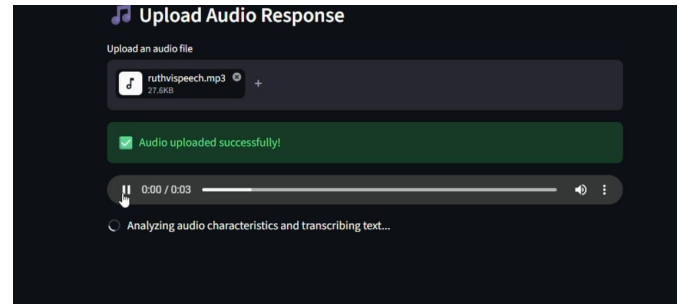


Figure 6: Audio Upload

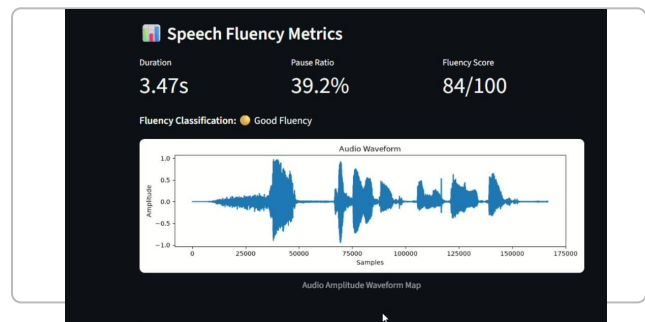


Figure 7: Audio Waveform

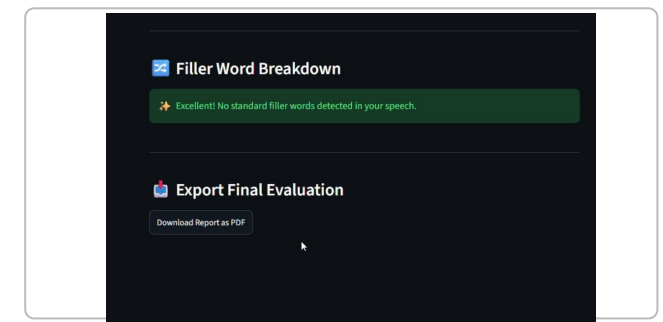


Figure 8: Filler word detection

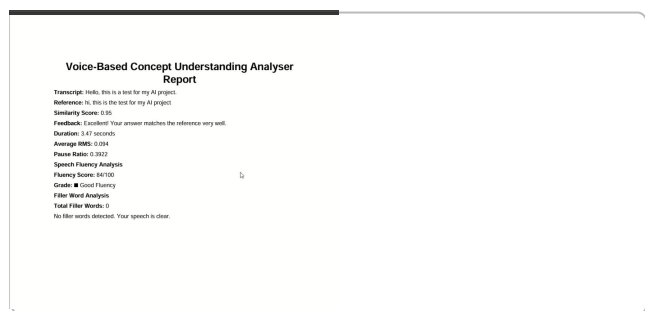


Figure 9: Generated PDF Report

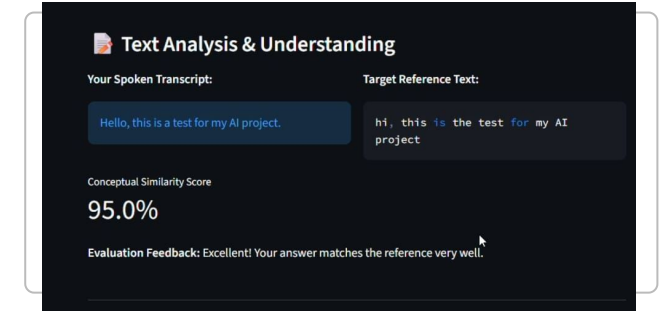


Figure 10: Strength & Semantic Analysis

6. Demonstration Summary

The application was successfully executed and tested. All major functionalities including audio upload, speech-to-text conversion, semantic similarity analysis, score generation,

waveform visualization, dashboard display, and PDF report generation worked successfully during the demonstration.